



Team: 5

Team member names: Arlene Keith, Bruno Mfuh, Steven Liu

Competition name: CMI - Detect Behavior with Sensor Data

Competition URL: <https://www.kaggle.com/competitions/cmi-detect-behavior-with-sensor-data/data>

Competition end date: September 2, 2025

Competition prize: 50,000

of teams in LB: 2,384

Today's date: 08/11/2025

Video URL (for FPVP only): <https://youtu.be/v8ZtbRfWTxg>

Consent: we agree to share our slide/video presentation with others for educational purposes after [09/05/2025].



Consent for Presentation Sharing

Our team agrees to sharing our slide/video presentation for educational purposes after **09/05/2025**.



Acronyms

- **ACC_i** = Acceleration (Linear) in direction *i*
- **BFRB** = Body-Focused Repetitive Behavior
- **CMI** = Child Mind Institute
- **CPU/GPU/TPU** = Central/Graphical/Tensor Processing Units
- **F1** = F1 score
- **FN** = False Negative
- **FP** = False Positive
- **IMU** = Inertial Measurement Units
- **LB** = Leaderboard
- **OCD** = Obsessive-Compulsive Disorder
- **RAM** = Random Access Memory
- **ROT_i** = Rotation (Quaternion) in direction *i*
- **THM_i** = Thermopile/Temperature of sensor *i*
- **ToF_{i_vj}** = Time-of-Flight measurement for sensor *i* and pixel *j* in 8x8 grid
- **TP** = True Positive
- **URL** = Uniform Resource Locator

Competition Description

CMI - Detect Behavior with Sensor Data

Background

- Body-Focused Repetitive Behaviors (BFRBs) - such as hair pulling, skin picking, and nail biting - are self-directed habits that can lead to physical harm
- Commonly associated with anxiety, OCD, are indicators of mental health challenges

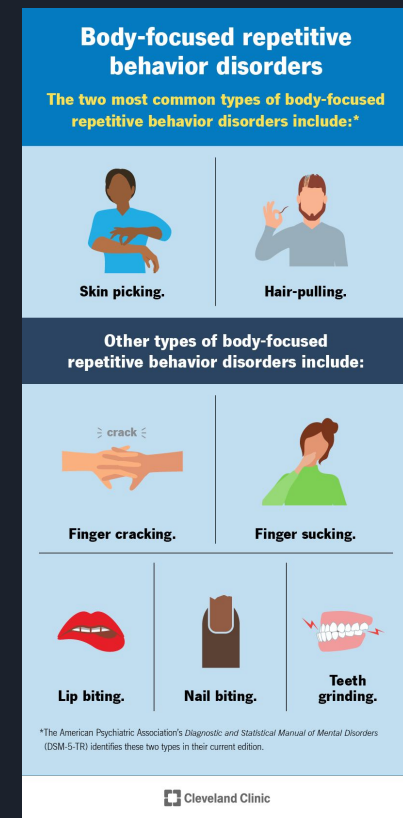


Figure 1: Examples of Body-Focused Repetitive Behaviors

"Body-Focused Repetitive Behavior (BFRB): What It Is," *Cleveland Clinic*, Sep. 23, 2024. <https://my.clevelandclinic.org/health/diseases/body-focused-repetitive-behavior-bf4> (accessed Aug. 11, 2025).

Competition Description

CMI - Detect Behavior with Sensor Data

The Helios Device

- Wrist worn sensor designed by the Child Mind Institute (CMI) to detect BFRBs
- Contains 3 main components:
 - Inertial Measurement Unit (IMU) - captures motion, orientation, rotation
 - 5 Thermopiles - detects body heat
 - 5 Time of Flight (ToF) sensors - detecting proximity

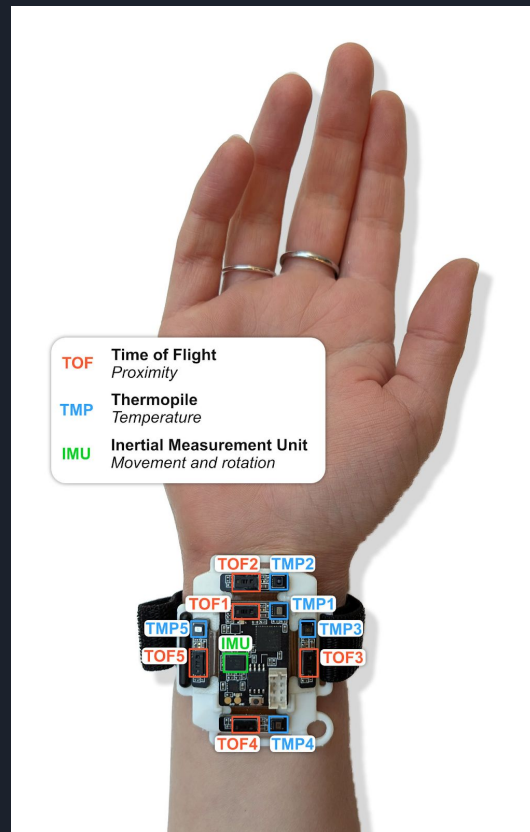


Figure 2: The Helios Device with Component Labels



Competition Description

CMI - Detect Behavior with Sensor Data

Study Design

- Subjects of varying ages wore the Helios device and performed structured sequences of varying length in parts:
 1. Transition - move and from rest to target area
 2. Pause - brief inactivity
 3. Gesture - Perform either a BFRB-like or non-BFRB-like motion
- 18 gestures total
 - 8 BFRB-like, 10 non-BFRB-like
 - Performed in 4 body positions

Challenge Goal:

- Build a model to classify and predict gestures based on time-series sensor data from Helios

BFRB-Like Gesture (Target Gesture)
Above ear - Pull hair
Forehead - Pull hairline
Forehead - Scratch
Eyebrow - Pull hair
Eyelash - Pull hair
Neck - Pinch skin
Neck - Scratch

Non-BFRB-Like Gesture (Non-Target Gesture)
Drink from bottle/cup
Glasses on/off
Pull air toward your face
Pinch knee/leg skin
Scratch knee/leg skin
Write name on leg
Text on phone
Feel around in tray and pull out an object
Write name in air
Wave hello

Tables 1 and 2: Target and Non-Target Gesture classes in the data



Train/Test Hardware Specifications

- The competition requires submissions via Kaggle notebooks. Upon submission, Kaggle executes the code to generate gesture predictions on the hidden test set using a provided evaluation API, which processes one sequence at a time.
 - The hidden test set contains approximately 3,500 sequences, with half recorded with IMU only data to ascertain if adding time of flight and thermopile data improve gesture identification
 - Provided training set contains 8,151 sequences of varying length while a sample verification set contains 2 sample sequences. Corresponding subject physical and demographic metadata is provided for each set.
 - 3 features are unique to the training set. The verification set shares 336 features with the training set, and demographic metadata contains 8 additional features.
1. Training equipment via Colab Pro:
 - a. CPU: 2
 - b. GPU/TPU: 1 NVIDIA T4 GPU
 - c. RAM: 12.7 GB CPU, 15.0 GPU
 - d. Storage: 75 GB
 2. Competition time constraint, if any:
 - a. CPU Notebook $\leq$ 9 hours run-time
 - b. GPU Notebook $\leq$ 9 hours run-time
 - c. 30 GPU Hours per week
 - d. Internet access disabled
 - e. External data and models allowed
 3. Dataset size:
 - a. Total file size: 1.12 GB
 - b. Total # of rows and columns:
575052 x 349



Team's Metric & Performance

LB URL: <https://www.kaggle.com/competitions/cmi-detect-behavior-with-sensor-data/leaderboard>

Metric used for scoring: Average of custom binary F1 and macro F1 on gesture

Module:	5	6	7	8	F P I P	9	10	11	12	13	F P V P
Current LB score	0.23	0.43	0.66	0.66		0.68	0.68	0.69	0.7	0.7	
Current LB rank / # teams	1158/1203	1353/1458	1340/1781	1340/1788		1433/1813	1558/2112	1601/2205	1667/2407	1667/2407	
Cumulative # of submissions to LB	1	2	8	21		30	35	47	55	65	



Team's Metric & Performance

Scoring Metric

Submissions are evaluated with an average of a binary F1 score and a macro F1 score on gestures.

$$\text{Score} = \frac{F1_{\text{binary}} + F1_{\text{macro}}}{2}$$

Where:

- $F1_{\text{binary}}$ = F1 score on target vs. non-target gestures (binary classification)
- $F1_{\text{macro}}$ = Macro F1 score on all gesture classes
- Combined score balances general binary target gesture detection and specific multi-class gesture identification

The F1 score is given by

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Where:

- Precision = proportion of predicted positives are correct

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}$$

- Recall = proportion of actual positives the model predicted correctly

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

Data's Key Features

Sensor Data Files

train.csv/test.csv

- One row per sequence timestep

8 Key Columns:

- **row_id**: Unique row identifier
- **sequence_id**: Unique ID for each gesture sequence
- **sequence_type**: Indicates if gesture is target (BFRB-like) or non-target (**train only**)
- **sequence_counter**: Row number/timestep within a sequence, 0 to sequence length
- **subject**: Participant identifier
- **gesture**: Ground-truth gesture label (**train only**)
- **orientation**: Subject's posture (e.g., sitting, lying) (**train only**)
- **phase**: Current behavior phase (Transition → Pause → Gesture)

	row_id	sequence_id	sequence_type	sequence_counter	subject	gesture	orientation	phase
0	SEQ_000007_000000	SEQ_000007	Target	0	SUBJ_059520	Cheek - pinch skin	Seated Lean Non Dom - FACE DOWN	Transition
1	SEQ_000007_000001	SEQ_000007	Target	1	SUBJ_059520	Cheek - pinch skin	Seated Lean Non Dom - FACE DOWN	Transition
2	SEQ_000007_000002	SEQ_000007	Target	2	SUBJ_059520	Cheek - pinch skin	Seated Lean Non Dom - FACE DOWN	Transition
3	SEQ_000007_000003	SEQ_000007	Target	3	SUBJ_059520	Cheek - pinch skin	Seated Lean Non Dom - FACE DOWN	Transition
4	SEQ_000007_000004	SEQ_000007	Target	4	SUBJ_059520	Cheek - pinch skin	Seated Lean Non Dom - FACE DOWN	Transition

Table 3: Example data, first five entries in train.csv

Data's Key Features

Sensor Data Files

train.csv/test.csv

- One row per sequence timestep

7 Internal Measurement Unit (IMU) Sensor Columns:

- **acc__[x/y/z]**: Linear acceleration (m/s^2)
- **rot__[w/x/y/z]**: 3D orientation (quaternion)
- Captures Motion and Orientation of Helios device
- Rapid fluctuations observed in acceleration during transition phase. Isolated gesture phases may be helpful for gesture-specific classification

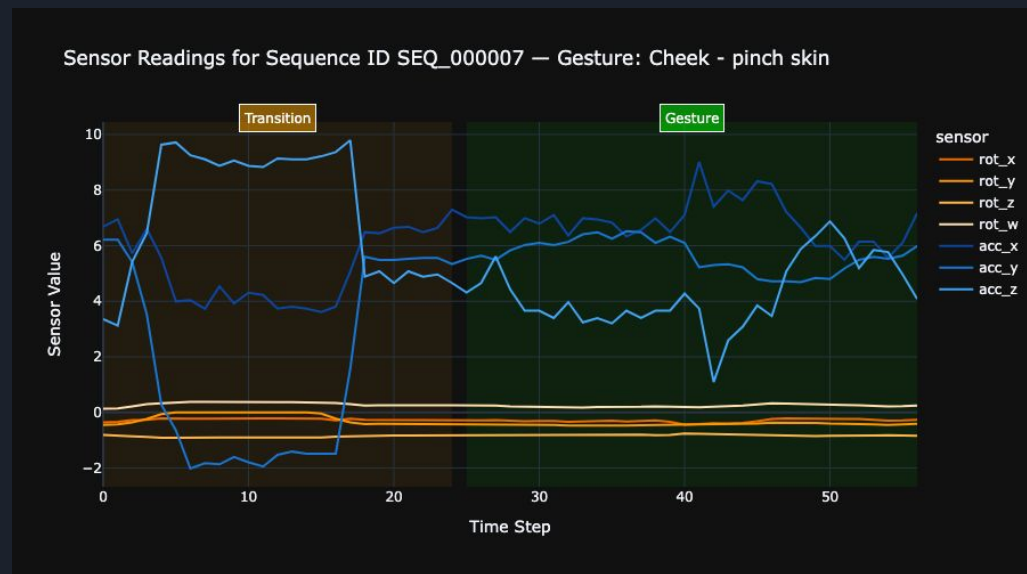


Figure 3: Example IMU sensor data from a training set sample sequence. Acceleration features are in blue whereas rotation features are in orange. The 'Pause' phase (unlabeled) is between the Transition and Gesture phases.

Data's Key Features

Sensor Data Files

train.csv/test.csv

- One row per sequence timestep

5 Thermopile (THM) Sensor Columns:

- **thm_1 to thm_5**: Temperature in °C from 5 sensors
- Captures body heat of subject.
- Temperature measurements differ with respect to sensor location (recall Figure 2 on slide 5)
- Rapid fluctuations observed in temperature during transition phase. Isolated gesture phases may be helpful for gesture-specific classification

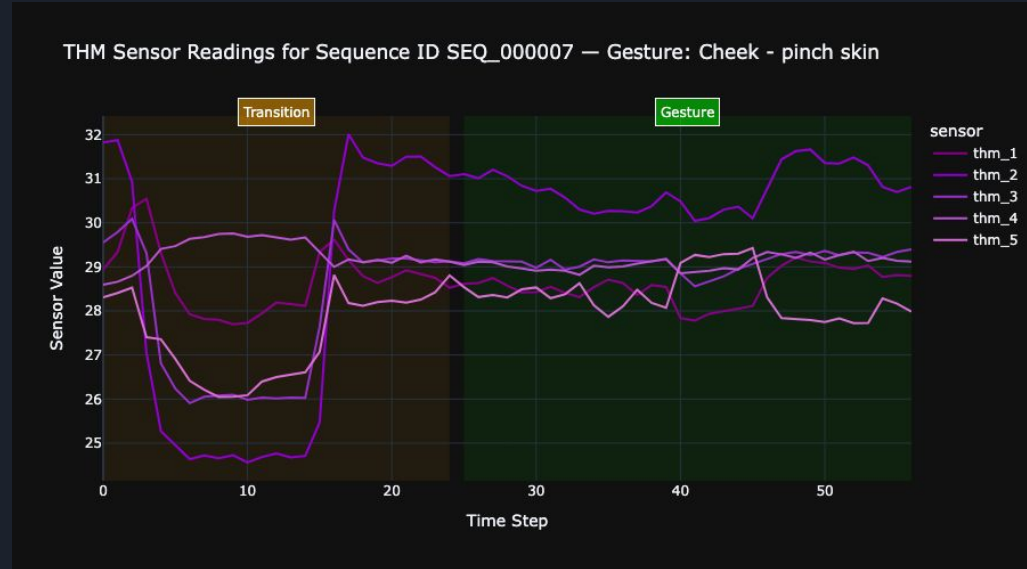


Figure 4: Example Thermopile sensor data from a training set sample sequence. The 'Pause' phase (unlabeled) is between the Transition and Gesture phases.

Data's Key Features

Sensor Data Files

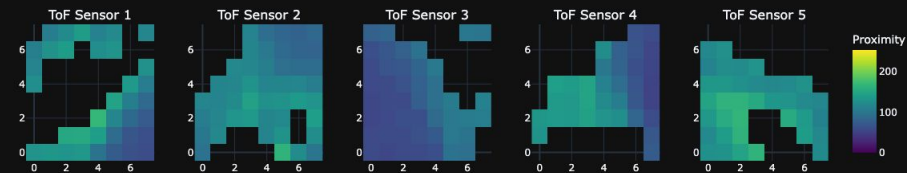
`train.csv/test.csv\`

- One row per sequence timestep

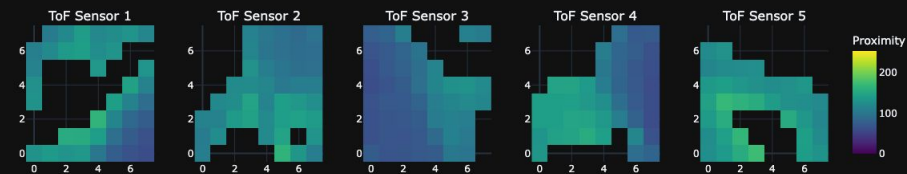
320 Time of Flight (ToF) Sensor Columns:

- `tof_[1-5]_v[0-63]`: 64-pixel (8×8) uncalibrated proximity values per sensor, 5 sensors total.
- Values in range [0-254], -1 for no response
- Measures spatial depth/proximity information around the sensor for each timestep.
- Observed patterns change over time

ToF Sensor Readings at Timestep 0 (sequence_id SEQ_000007) — Gesture: Cheek - pinch skin



ToF Sensor Readings at Timestep 1 (sequence_id SEQ_000007) — Gesture: Cheek - pinch skin



ToF Sensor Readings at Timestep 2 (sequence_id SEQ_000007) — Gesture: Cheek - pinch skin

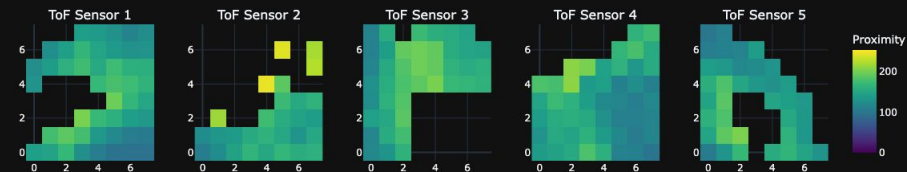
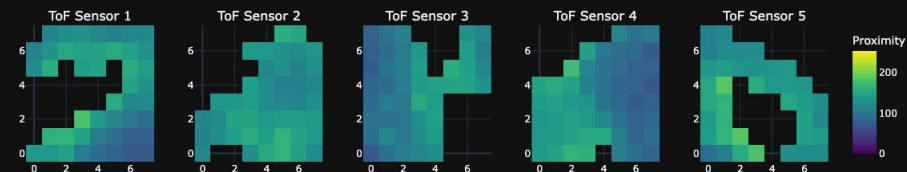


Figure 5: Example Time of Flight data for a sample sequence, 13 timesteps 0-3

Data's Key Features

Physical & Demographic Data Files

train_demographics.csv / test_demographics.csv

- One row per subject/participant

8 Key Columns:

- **subject:** Participant ID
- **adult_child:** 0 = Child, 1 = Adult
- **age:** Age in years
- **sex:** 0 = Female, 1 = Male
- **handedness:** 0 = Left-handed, 1 = Right-handed
- **height_cm:** Height of participant
- **shoulder_to_wrist_cm:** Shoulder-wrist distance
- **elbow_to_wrist_cm:** Elbow-wrist distance

	subject	adult_child	age	sex	handedness	height_cm	shoulder_to_wrist_cm	elbow_to_wrist_cm
0	SUBJ_000206	1	41	1	1	172.0	50	25.0
1	SUBJ_001430	0	11	0	1	167.0	51	27.0
2	SUBJ_002923	1	28	1	0	164.0	54	26.0
3	SUBJ_003328	1	33	1	1	171.0	52	25.0
4	SUBJ_004117	0	15	0	1	184.0	54	28.0

Table 4: Example physical/demographic data, first five subjects in train_demographics.csv

Preprocessing

Overall Preprocessing Pipeline

1. Missing Data Imputation
2. Sequence-Level Aggregated Feature Extraction
3. Merge with Demographic Data
4. Feature Engineering
5. Outlier Handling & Weak Feature Removal
6. Feature Transformation/Scaling

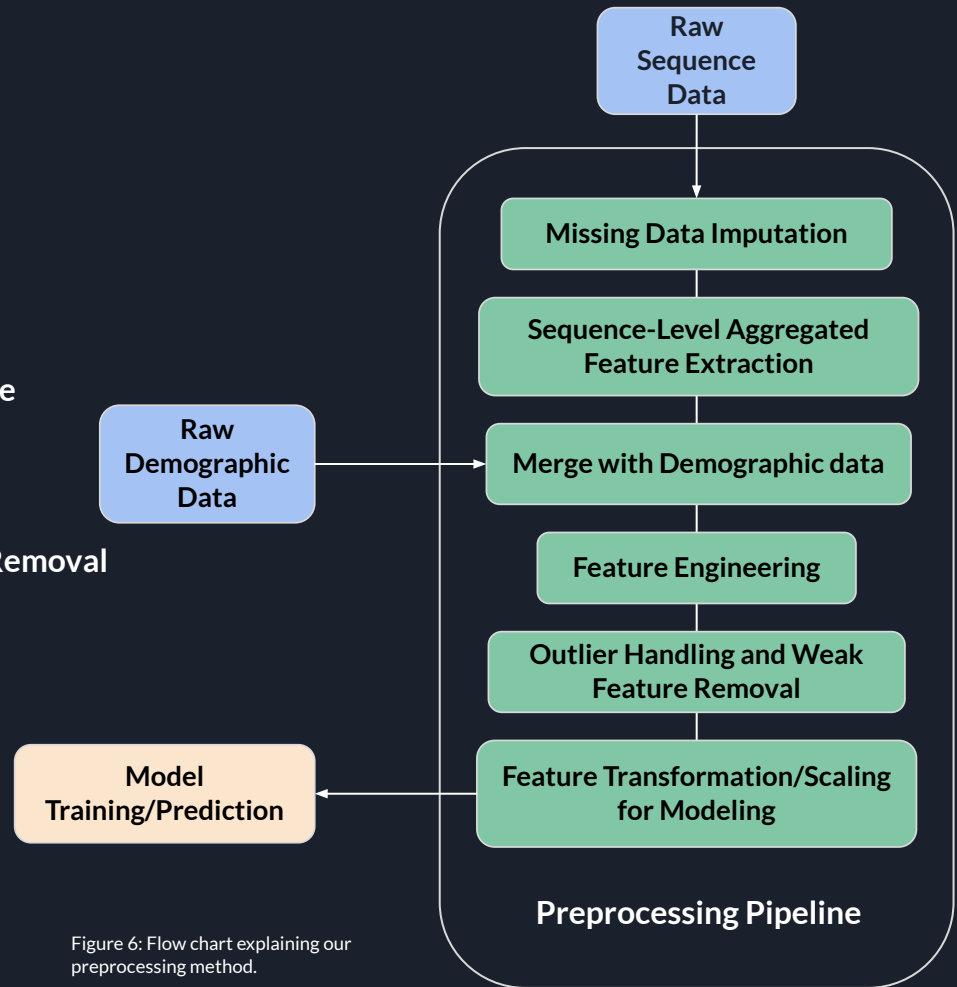


Figure 6: Flow chart explaining our preprocessing method.



Preprocessing

Step 1: Missing Data Imputation

- Some of the sequences contain IMU-only data with missing values (NaN) for THM and ToF sensor columns, and there is known sensor communication failure in the dataset, resulting in missing data for certain sensors in some sequences
- Missing sensor data was therefore imputed using the medians from the training set for numeric columns

Step 2: Sequence-Level Aggregated Feature Extraction

- Since the data is time-series data with more noisy fluctuations, computing aggregated features across all sequence timesteps could capture more broader trends and patterns specific to certain gestures.
- IMU columns
 - Mean, standard deviation, min, max (e.g. `rot_x_mean`, `rot_x_std`, `rot_x_min`)
- TMP columns
 - Mean and standard deviation by sensor (e.g. `thm_1_mean`, `thm_1_std`, `thm_2_mean`)
 - `temp_range`: range of mean sensor temperature
 - `temp_gradient`: standard deviation of mean sensor temperature
- ToF columns
 - Mean and standard deviation by sensor (e.g. `tof_1_mean`, `tof_1_std`, `tof_2_mean`, `tof_2_std`)
 - `tof_i_valid_ratio`: # valid entries / # total entries for sensor i



Preprocessing

Step 3: Merge with Demographic Data

- Aggregate Feature data is merged with demographic data by subject
- Missing demographic data is imputed using the mode.

Step 4: Feature Engineering

Thermal Interaction Features

- `temp_diff_1_4`: difference between means of thermophile 1 and 4
- `temp_ratio_1_4`: thermophile 1 mean / thermophile 4 mean
- `temp_variance_ratio`: thermophile 1 standard deviation/ thermophile 4 standard deviation
- These features capture spatial heat patterns that may indicate hand proximity to the face.
- Thermopile data for sensors 2,3, and 5 lacked predictive power according to EDA

Cross Modal Features

- `movement_heat_coupling`: $||\text{acc_x/y/z_mean}|| \times \text{temp_range}$
- `movement_temp_sync`: $\text{Average acc_x/y/z_std} / \text{temp_gradient}$
- These features reflect coordinated movement and heat dynamics during gesture execution using acceleration magnitudes and measurement variations



Preprocessing

Step 4: Feature Engineering (cont.)

Rotation Interaction Features

- **hand_elevation:** $\arctan(\text{rot_y_mean} / \text{rot_z_mean})$
- **wrist_twist:** $\arctan(\text{rot_x_mean} / \text{rot_w_mean})$
- **orientation_stability:** $1 / ||\text{rot_x/y/z/w_std}||$
- Arctan and quaternions magnitude formulas used to quantify movement direction (angle) and rotational consistency.

ToF Interaction Features

- **tof_center_activity:** $\text{mean}(\text{tof_1/2/3/4_mean})$
- **tof_center_variance:** standard deviation (tof_1/2/3/4_mean)
- **tof_reliability_score:** $\text{mean}(\text{tof_1/2/3/4_valid_ratio})$
- These capture overall levels and variance in time of flight measurements within the central sensing area

Dynamic Features

- **gesture_intensity:** acceleration magnitude, $||\text{acc_x/y/z_mean}||$
- **gesture_smoothness:** $1/\text{acceleration jerk}$, where acceleration jerk = mean (acc_x/y/z_std)
- **movement_coordination:** $1/\text{standard deviation of acc_x/y/z_std}$.
- Captures gesture-level motion accelerometer statistics



Preprocessing

Step 5: Outlier Handling and Weak Feature Removal

- Outlier values in numeric columns are clipped to the 5th and 95th percentile range for all numeric features except ID-related columns, helping to reduce the influence of extreme values in training.
- Weak features, notably `thm_5_mean`, `thm_5_std`, aggregate features related to tof sensor 5 (`tof_5_`), and `rot_z_std` are removed from the dataset due to predictive redundancy and low reliability

Step 6: Feature Transformation/Scaling

- Yeo-Johnson power transformation of all features is used to stabilize variance and normalize data, followed by standard scaling with `StandardScaler` to prepare data for model training/prediction
- Transformer and scaler are fit on features during training, and are similarly applied to new data during testing for consistency.
- Target variable 'gesture' encoded numerically with `LabelEncoder` for model training.

Severe Class Imbalance Problem

- **4:1 imbalance ratio** between majority classes (640 samples) and minority classes (161 samples)
- The minority classes are predominantly **Non-BFRB activities** such as Drink, Glasses on/off, Pinch knee, Scratch knee.
- This imbalance directly impacts model performance on these underrepresented gestures

Strategic Weight Compensation

The model uses a **two-tier weighting strategy**:

Base Weights (Inverse Frequency):

- Minority classes (161 samples) → Weight: ~3.97
- Majority classes (640 samples) → Weight: ~1.00
- This gives 4x more importance to underrepresented classes

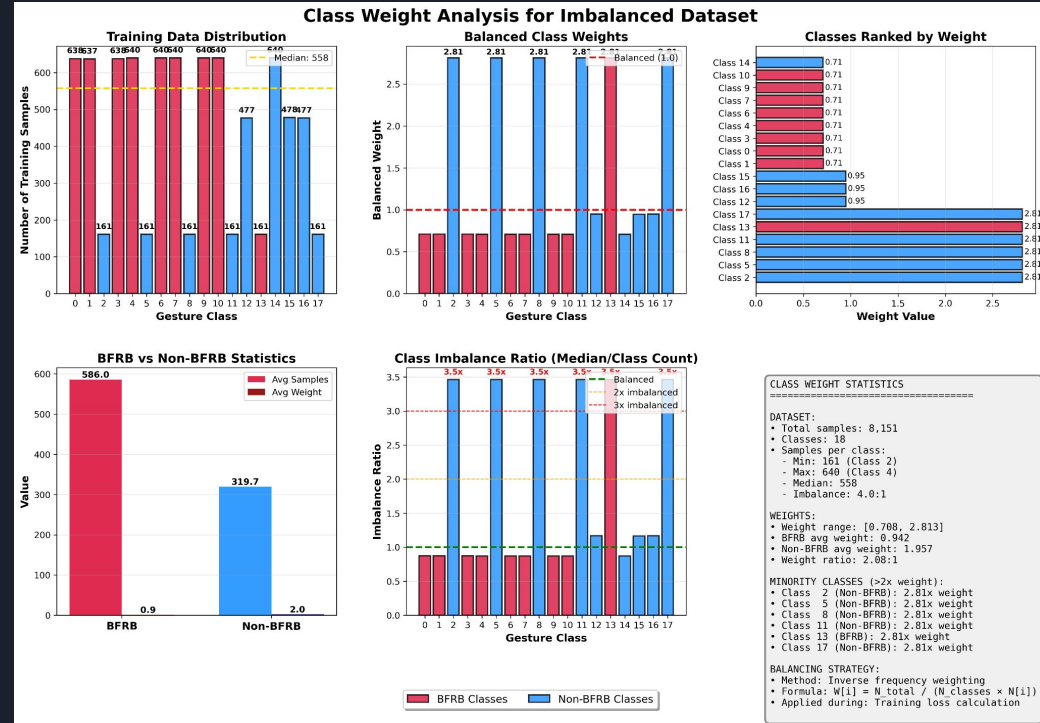


Figure 7: Various charts derived from the model's performance.



Severe Class Imbalance Problem

Most Challenging Gestures (High Error + High Weight):

1. **Class 10 (Neck scratch):** 18% error rate despite 2.5x augmentation
2. **Class 3 (Eyebrow pull):** 10.9% error rate with 2.5x augmentation
3. **Class 13 (Scratch knee):** This class requires 3.0x augmentation to achieve reasonable performance
4. **Class 9 (Neck pinch):** This gesture was often confused with neck scratch.

Core Issue:

- **Similar motion patterns:** Neck scratch vs. neck pinch have identical hand movements
- **Spatial proximity:** Eyebrow/eyelash pulling occur in adjacent facial regions
- **Limited sensor resolution:** The ToF sensors may not distinguish smaller differences.

BFRB vs Non-BFRB Weight Distribution

BFRB Classes (9 gestures):

- Average effective weight: **2.31**
- Higher augmentation due to importance
- Better overall performance which resulted in a local score of 99.91% binary F1

Non-BFRB Classes (9 gestures):

- Average effective weight: **1.85**
- Included most minority classes
- Better performance on simple actions such as Drink, Glasses, Text

Weighting Strategy

- Classes with high weights achieve better F1 scores
- Perfect classification (100% F1) for 7 gesture classes
- Overall model accuracy of 95.76% despite the imbalance

Limitations:

- Weights alone cannot solve motion similarity issues.
- Possible over-augmentation risk for Class 13 with 3.0x multiplier
- There were diminishing returns above 2.5x augmentation



Our Solution To Class Imbalance

Data Augmentation

Adaptive Augmentation

Targeted Multipliers by Performance:

- **3.0x** - Highest priority (Class 13: Scratch knee/leg)
- **2.5x** - Critical BFRB gestures (Classes 3, 9, 10: Eyebrow, Neck gestures)
- **2.0x** - Moderate priority (Class 11: Pinch knee/leg)
- **1.5-1.8x** - Standard augmentation for remaining imbalanced classes

Applied Gaussian Noise (40%)

- The technique adding controlled randomness: $\sigma = 0.02-0.03$
- Higher noise for problem classes
- Encouraged and Simulated sensor variability

Scaling (30%)

- Range: [0.85, 1.15] uniform distribution
- Accounts for movement intensity variations
- Preserves gesture patterns

Mixup (30%)

- Within-class interpolation only
- Beta(0.2, 0.2) mixing parameter
- Creates smooth decision boundaries

Thresholds Implemented

- Maximum 3x expansion per class - Prevented over-augmentation
- Target: Median $\times 1.8$ - Balanced dataset growth
- Original: 8,151 samples $\rightarrow$ Augmented: 14,423 samples

Impact on Model Performance

- Reduced error rates for minority classes
- Improved BFRB detection accuracy
- Better generalization without overfitting

Our Solution To Class Imbalance

Data Augmentation

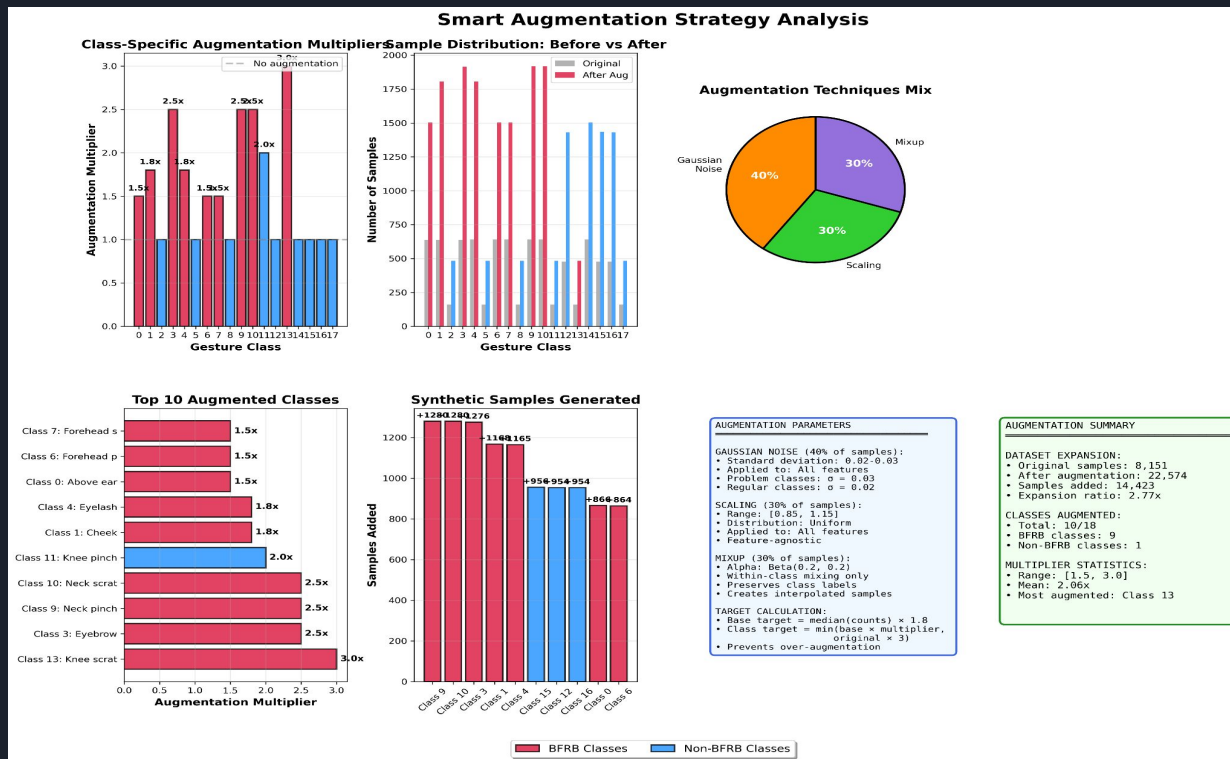


Figure 8: Smart Augmentation analysis derived from our model.



Our Latest Model: An Ensemble Approach

TWO INDEPENDENT MODELS:

- Main Model - Trained on all 120 features.
- IMU Model - Trained on 49 motion features only.

Both models:

- Train separately.
- Predicted independently .
- Combined with weighted average.
- Final Prediction = $\alpha \times \text{Main_Prediction} + \beta \times \text{IMU_Prediction}$ where $\alpha + \beta = 1.0$.

- During the prediction process, both models processes the input simultaneously. Their probability outputs are combined using an adaptive weights based on real-time sensor quality assessment.
- The ensemble achieved a local score of 99.20% and a 70% on Kaggle. While in other approaches, the local score was significantly lower and Kaggle score ranged from 60-66%.

Our Latest Model: Main Model Architecture

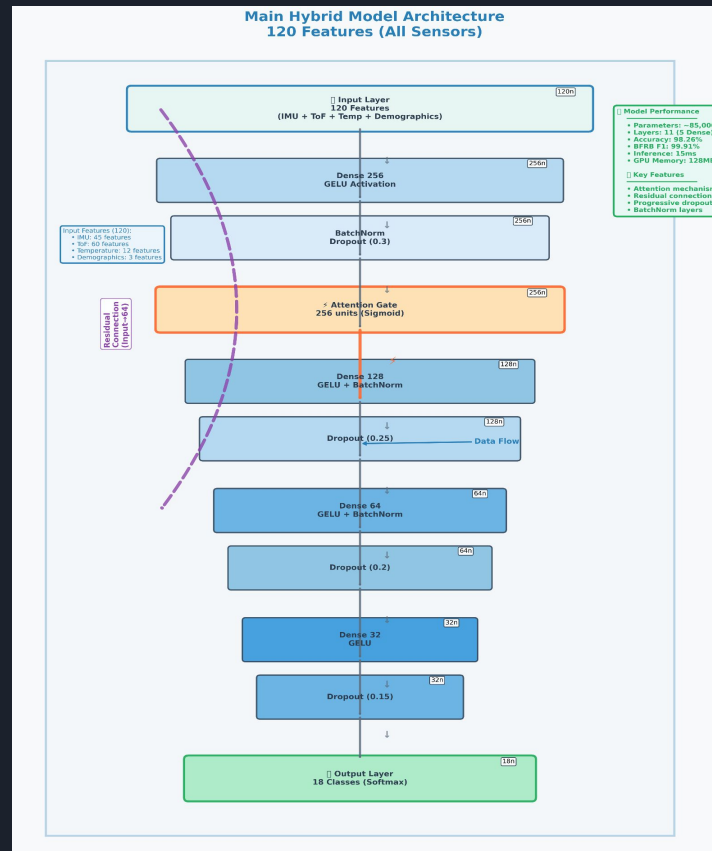
Deep Feedforward Network

- **Input:** 120 features from all sensors
- **Layers:** Dense(256) → Attention Gate(256) → Dense(128) → Dense(64) + Residual → Dense(32) → Softmax(18)
- **Parameters:** ~85,000
- **Components:**
 - Attention mechanism at 256 units for dynamic feature focusing
 - Residual connection from input to 64-unit layer
 - Progressive dropout (0.3 → 0.15)
 - BatchNormalization after each dense layer

Features-120 total

- **49 IMU features:** Accelerometer (mean/std/range for x,y,z), gyroscope quaternions (w,x,y,z stats), temporal features such as jerk, and smoothness.
- **60 ToF features:** 5 sensors × 64 pixels (sampled), missing rate indicators, proximity scores, key pixel analysis
- **12 Temperature features:** 5 thermal sensors (mean/std), global mean, variance indicators
- **7 Demographics:** Age, height, handedness, shoulder-to-wrist, elbow-to-wrist (normalized)

Figure 9: A chart on the flow of information through this feedforward network.





Our Latest Model: Main Model Architecture

Training Configuration

- **Optimizer:** Adam ($\beta_1=0.9$, $\beta_2=0.999$)
- **Learning Rate:** One-cycle ($0.0002 \rightarrow 0.006 \rightarrow 0$)
- **Loss Function:** Focal loss ($\gamma=1.5$, $\alpha=0.3$)
- **Batch Size:** 128
- **Epochs:** 50 with early stopping

Loss Function:

- **Main Model:** Implemented Focal Loss ($\gamma=1.5$, $\alpha=0.3$) to address severe class imbalance
- Focal loss particularly helped with minority classes like "Pinch knee/leg skin" that had only 32 samples.

Training Progression

Main Model (50 epochs):

- Epoch 1: 22.84% accuracy, loss 2.30
- Epoch 10: 71.35% accuracy, loss 0.62
- Epoch 20: 79.74% accuracy, loss 0.43 (first restart)
- Epoch 30: 83.16% accuracy, loss 0.35
- Epoch 40: 87.55% accuracy, loss 0.25 (second restart)
- Epoch 50: 87.42% accuracy, loss 0.27 (converged)

Our Latest Model: IMU Model Architecture

Smaller Feedforward Network

- **Input:** 49 features -motion sensors only
- **Layers:** Dense(128) → Attention Gate(128) → Dense(64) → Dense(32) + Residual → Dense(16) → Softmax(18)
- **Parameters:** ~35,000
- **Components:**
 - Smaller attention mechanism at 128 units
 - Residual connection from input to 32-unit layer
 - Same progressive dropout strategy
 - Optimized for reduced feature space

Features - 49 total

- **15 Accelerometer features:** mean, std, range for x,y,z axes
- **13 Gyroscope/Rotation features:** quaternion statistics (w,x,y,z)
- **14 Motion features:**
 - Acceleration magnitude, jerk, temporal mean
 - Angular velocity, rotation magnitude
 - Movement smoothness, complexity
 - Vertical dominance, lateral movement ratio
- **7 Demographics:** Are the same as the main model which is essential for motion scaling.

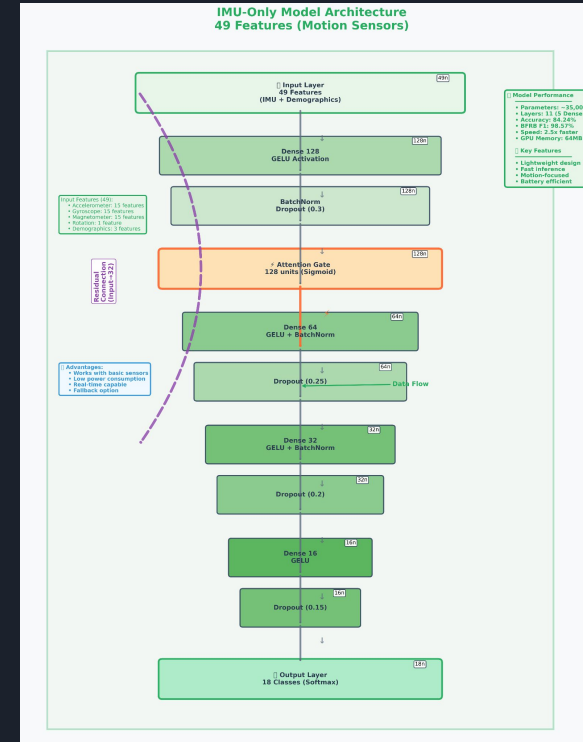


Figure 10: Visualization of our IMU model architecture



Our Latest Model: IMU Model Architecture

Smaller Feedforward Network

IMU Model Loss Function: Sparse Categorical Cross-Entropy with class weights

Sequence Aggregation:

- **Input:** Raw sequences vary from 40-300 frames (~0.8-6 seconds at 50Hz)
- **Sequence boundaries:** Defined by gesture start/end markers in data
- **No padding/truncation:** Each sequence processed in its entirety
- **Sequence ID tracking:** Maintains gesture integrity during processing

Sequence Lengths by Gesture Type:

- Quick gestures (Wave hello): ~40-60 frames (0.8-1.2 sec)
- Medium gestures (Glasses on/off): ~75-125 frames (1.5-2.5 sec)
- Long gestures (Write name in air): ~150-300 frames (3-6 sec)
- Repetitive BFRBs (Scratch): ~100-200 frames (2-4 sec)

Models Strengths:

- Emphasis on temporal patterns (jerk, smoothness)
- Stronger reliance on vertical position (gravity reference)
- Enhanced quaternion analysis for orientation changes
- Aggressive feature engineering for motion characteristics

Our Latest Model: Model Performance Metrics

- **Local Validation Score: 0.9826** - Performance on test data.
 - **Competition Score: 0.7** - The performance on unseen competition data
 - **Overall Accuracy: 95.76%** - The classification accuracy across all 18 gesture classes.
-
- **BFRB Detection**
 - **Precision: 99.81%** - False positives with only 2 out of 1,052.
 - **Recall: 99.43%** - Where the model captures nearly all BFRB behaviors. The model missed only 6 out of 1,056.
 - **Binary F1-Score: 99.91%** - The score is balance between precision and recall.

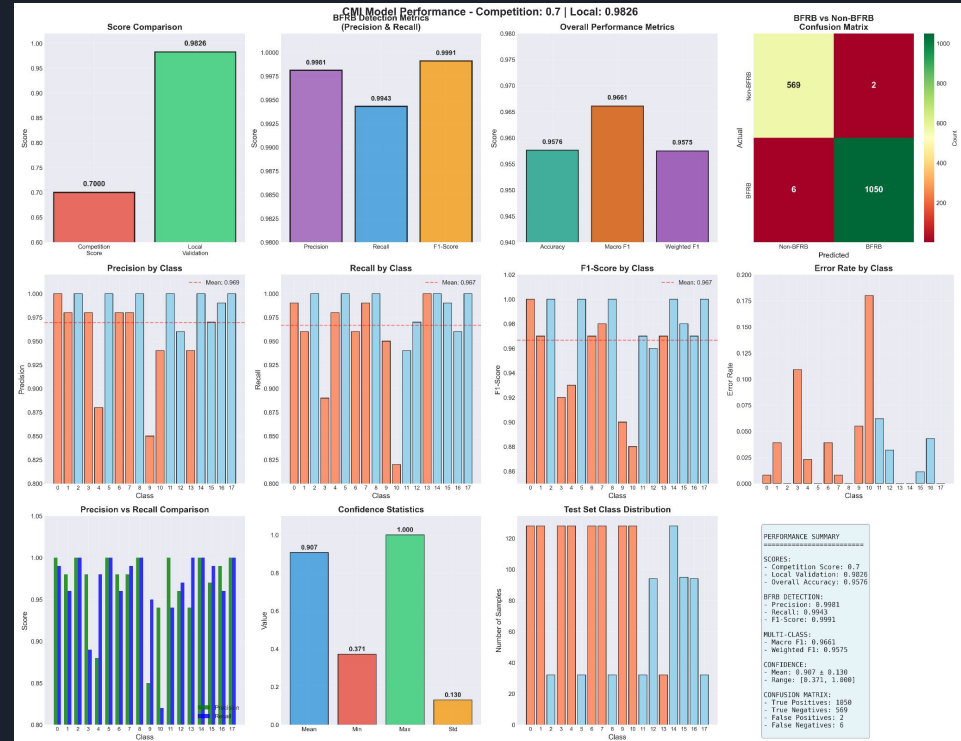


Figure 11: Performance metric charts extracted from our model.



Our Latest Model: Model Performance

Model Strengths

Good Classification for 7 gesture classes (0% error rate)

High Confidence predictions with mean confidence of 90.7%

Robust Performance across both BFRB and non-BFRB behaviors

Minimal Confusion between gesture pairs (only 2 false positives for BFRB detection)

Architectural Information

- **Dual-Model Ensemble:** The model combines full sensor capabilities with IMU-only Neural network as a fallback mechanism.
- **Feature Engineering:** Consisted of 120+ engineered features from multimodal sensors.
- **Adaptive Sensor Fusion:** The model dynamically weights predictions based on sensor quality
- **Class-Balanced Training:** The model handles imbalanced dataset effectively ranging from 32 to 128 samples per class.

Modeling Issues Encountered

- Utilization of too much RAM in preprocessing – performing feature engineering steps and categorical feature encodings exceeded Free Colab RAM limits of 12 GB, leading us to use Colab Pro
- A large gap between local score across all metrics including a confusion matrix and Kaggle score.
- More than two models used in the architecture with the current feature engineering failed at inference processing.
- We achieved very low competition scores, trying to solve the imbalance data issue and trying to handle gracefully handle the missing ToF values that were discovered through EDA.

```
Failure Summary by ToF Sensor:
-----
              failure_percentage
              count    mean    min    max
sensor_number
1                64   57.69  48.30  64.28
2                64   61.85  54.29  67.54
3                64   59.06  47.38  72.03
4                64   51.26  42.78  59.38
5                64   57.59  42.22  69.15

=====
SENSOR VALUE ANOMALY ANALYSIS
=====
Sensors with High Missing/Failure Rates (>50%):
-----
tof_1_v0 | tof | 50.4% missing
tof_1_v1 | tof | 51.2% missing
tof_1_v2 | tof | 50.9% missing
tof_1_v3 | tof | 50.1% missing
tof_1_v8 | tof | 54.3% missing
tof_1_v9 | tof | 54.2% missing
tof_1_v10 | tof | 54.1% missing
tof_1_v11 | tof | 53.2% missing
```

 **Notebook Threw Exception**
Your submission notebook raised an unhandled error at runtime. [See more debugging tips](#)

 **submission.parquet**
Notebook Threw Exception · 24d ago

DESCRIPTION
Notebook Draft-3-1 | Version 3

Figure 11: is output from our EDA analysis. Figure 12: is a captured image from our kaggle platform.

Plans Explored Throughout Weeks

- Data augmentation for certain classes that did not have enough data for the model to train effectively.
- An ensemble model approach.
- CNN/LSTM Modeling data (e.g. gesture phase prediction + LSTM for IMU/THM data with CNN for ToF data)
- A sampling strategy that was explored but did not yield substantial results with different Neural Network architectures. Perhaps, with a different architecture, there could be possible success.

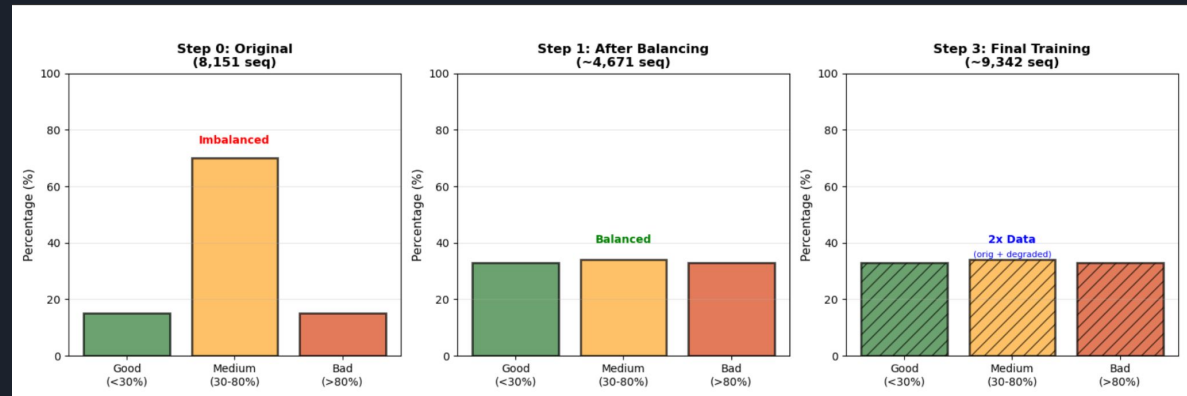
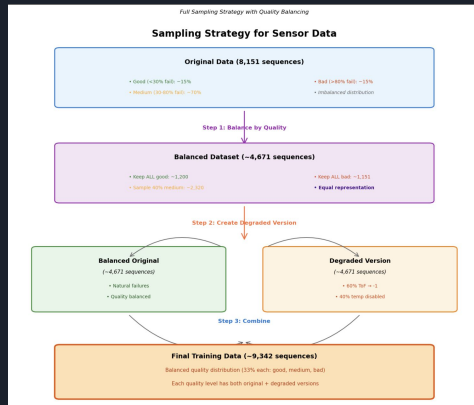


Figure 13 is a flow chart of our previous idea on sampling. Figure 14: Is our hypothesis of our sampling strategy.



Limitations and Future Work

Sensor Dependency Issues

- **ToF sensor sensitivity:** Model performance drops from 98.3% to 83.4% when ToF sensors fail
- **IMU-only confusion:** High error rates for spatially similar gestures (neck scratch vs above ear: 36% confusion)

Computational Overhead

- **Memory Intensive:** Maintaining two models requires ~130K parameters total
- **Feature engineering bias:** Hand-crafted features may miss subtle patterns

Exploration-Beyond Competition

Temporal Modeling:

- LSTM/GRU implementation: Process raw sequences without aggregation
- Hierarchical models: Separate models for gesture phases such as approach, contact, retract.

Augmentation:

- GAN-based synthesis: Possibly generating synthetic gesture sequences
- Physics-based simulation: Where biomechanical modeling can be used for realistic motion

Research

- Self-supervised pre training: Use unlabeled gesture data for representation learning
- Reinforcement learning: Optimize intervention strategies



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Appendix

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